

Supplementary information for:

Chapter 4 | Genomic Assessment of Inbreeding and Genetic Load in the Eurasian Curlew *Numenius arquata*

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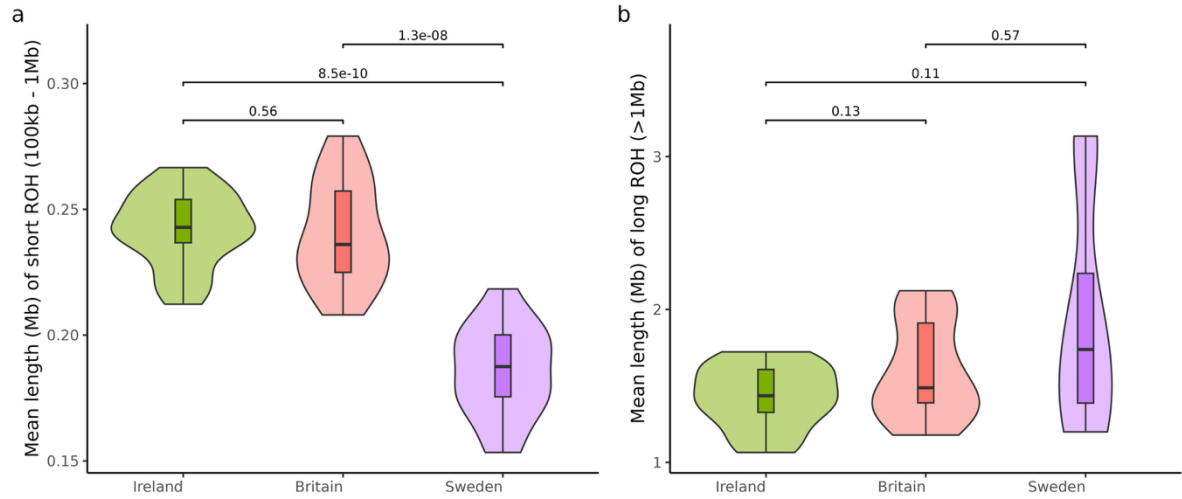


Figure S4.1: Mean length (Mb) of short (100 kb to 1 Mb) and long (> 1 Mb) runs of homozygosity (ROH) for three breeding populations of Eurasian curlew (*Numenius arquata*) in Ireland, Britain, and Sweden.

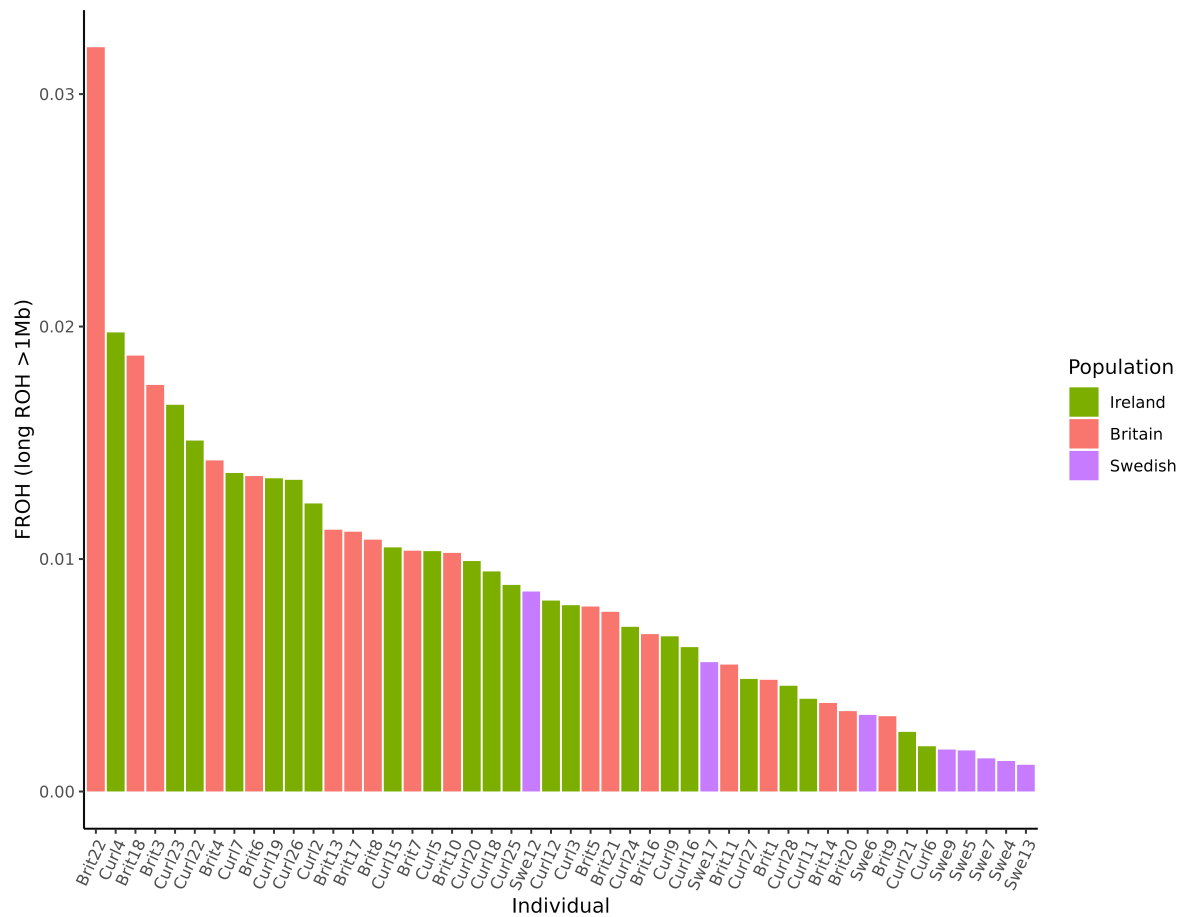


Figure S4.2: Distribution of pre-individual F_{ROH} , coloured by breeding population, for three Eurasian curlew breeding populations in Ireland, Britain, and Sweden.

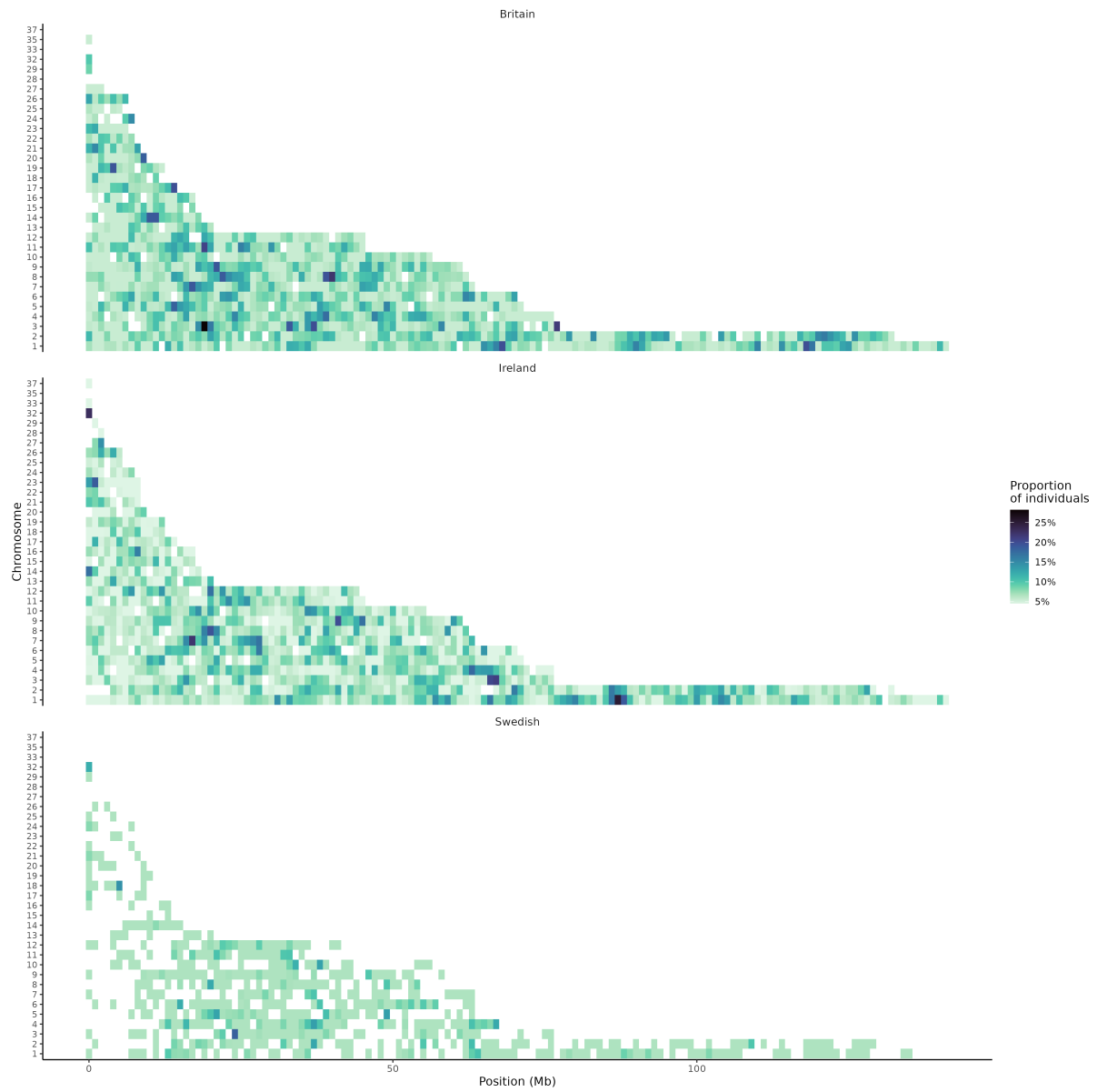


Figure S4.3: Heatmaps showing the proportion of individuals carrying runs of homozygosity (ROH) per genomic window across chromosomes for three breeding populations of Eurasian curlew (*Numenius arquata*) in Ireland, Britain, and Sweden

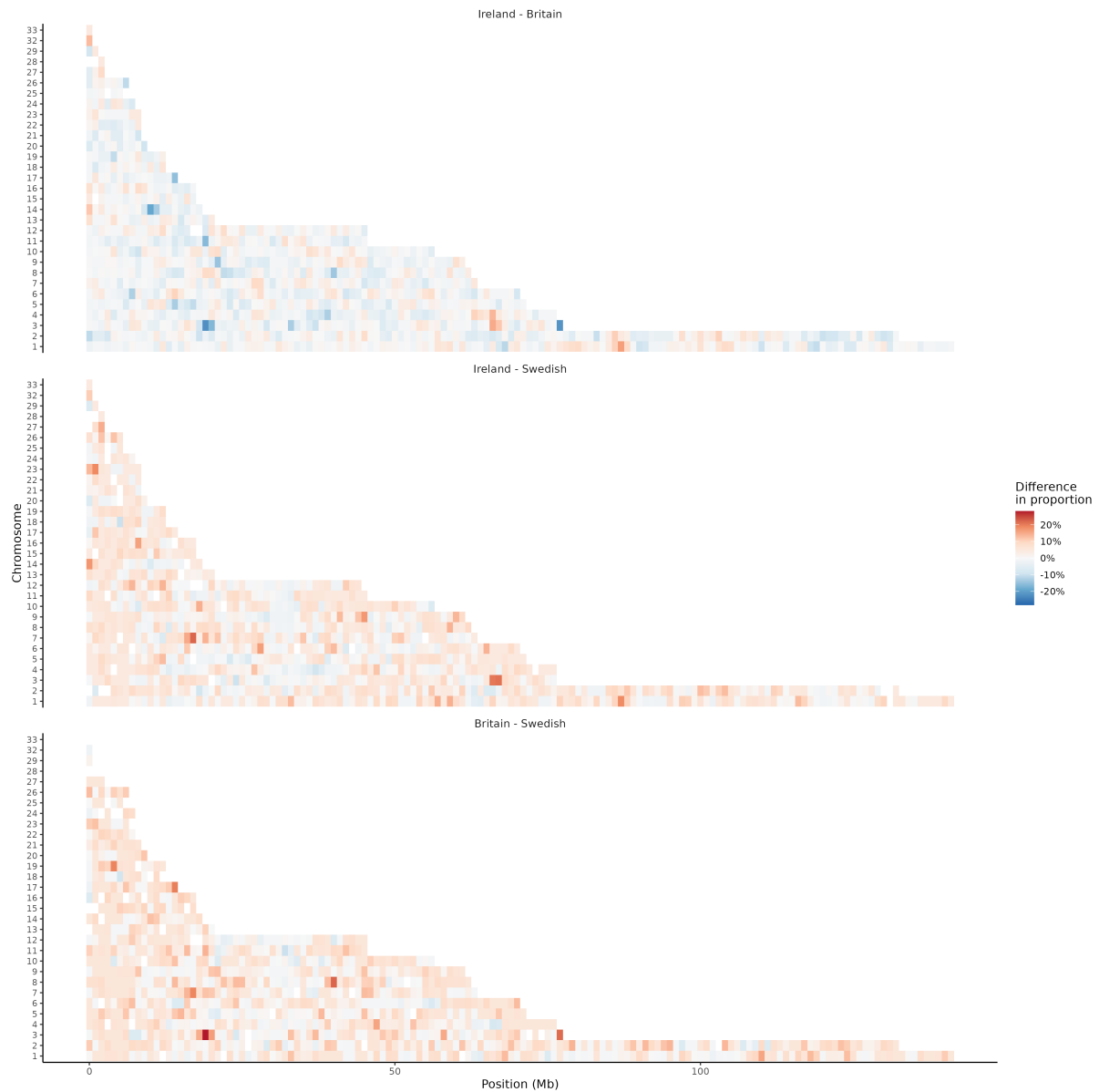


Figure S4.4: Heatmaps of runs of homozygosity showing differences in proportion across population comparisons of Eurasian curlew (*Numenius arquata*) in Ireland, Britain, and Sweden for 33 chromosomes. In the Ireland versus Britain and Ireland versus Sweden comparisons, positive values indicate a greater proportion of ROH in Ireland; in the Britain versus Sweden comparison, positive values indicate a greater proportion in Britain.

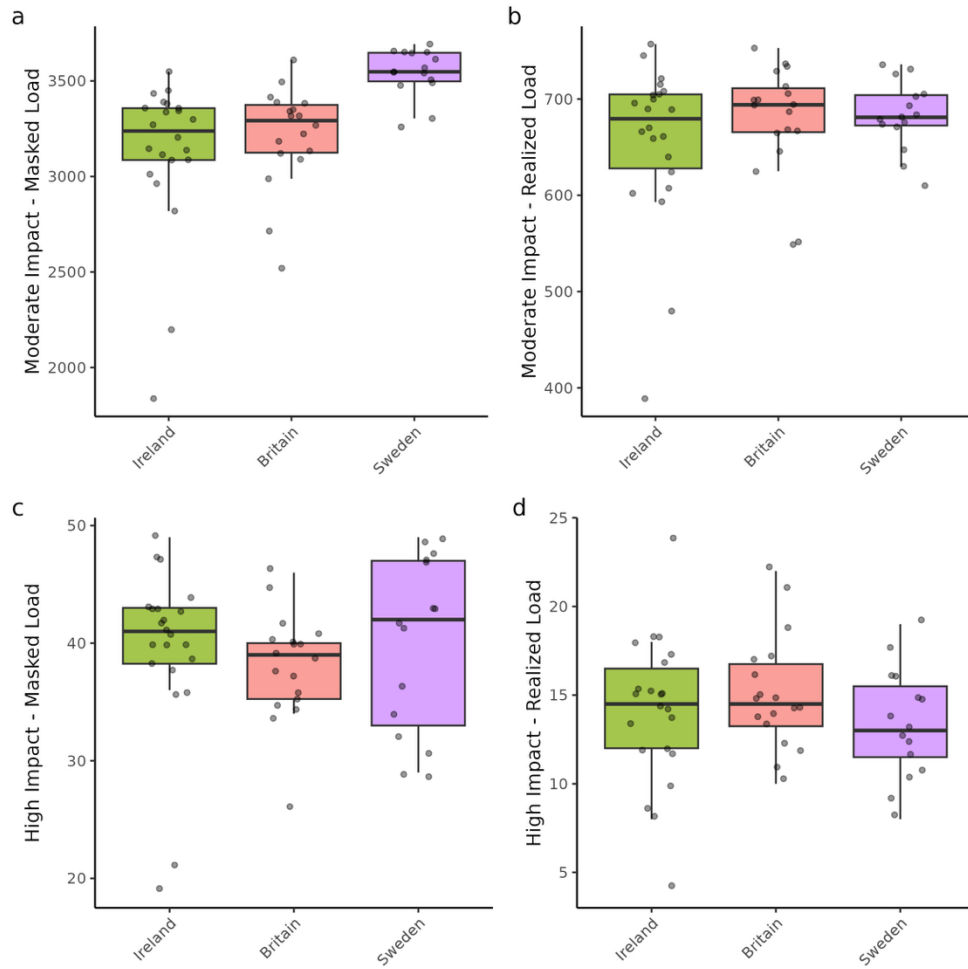


Figure S4.5: Raw counts of variants predicted to be deleterious partitioned by impact category and zygosity across three Eurasian curlew populations. (a) Moderate-impact masked load (heterozygous). (b) Moderate-impact realised load (homozygous). (c) High-impact masked load (heterozygous). (d) High-impact realised load (homozygous). Individual values are shown as points overlaid on boxplots. Variants were annotated using SnpEff.



Figure S4.6: Venn diagrams showing the shared and private variants predicted to have (a) moderate impact and (b) high impact in Eurasian curlews breeding in Ireland, Britain and Sweden identified with SnpEff.

Table S4.1: Kruskal-Wallis and pairwise Wilcoxon rank-sum test results (Benjamini-Hochberg corrected) for ROH metrics across Irish, British, and Swedish breeding populations.

Metric	KW_chi_sq	KW_p	Wilcoxon rank-sum <i>p</i> -values		
			Ire_vs_Brit	Ire_vs_Swe	Brit_vs_Swe
$F_{\text{ROH_short}}$	32.146	< 0.001	0.946	< 0.001	< 0.001
$F_{\text{ROH_long}}$	12.822	0.002	0.737	< 0.001	< 0.001
$F_{\text{ROH_total}}$	32.146	< 0.001	0.946	< 0.001	< 0.001
$N_{\text{ROH_short}}$	32.179	< 0.001	0.967	< 0.001	< 0.001
$N_{\text{ROH_long}}$	17.506	< 0.001	0.826	< 0.001	< 0.001
Mean length short	31.284	< 0.001	0.563	< 0.001	< 0.001
Mean length long	3.844	0.146	0.199	0.199	0.567

Table S4.2: Genotype counts and load fractions for the two outlier genes with elevated realised load, *DDO* and *MBOAT2*, across the three curlew populations. For details of the other identified genes see **Table S4.4**.

Gene	Pop	Het	Hom anc	Other	Hom der	<i>n</i>	Masked	Realised
<i>DDO</i>	Ireland	6	32	1	5	43	0.140	0.116
	Britain	6	26	1	3	35	0.171	0.086
	Sweden	7	14	1	8	29	0.241	0.276
<i>MBOAT2</i>	Ireland	7	9	0	6	22	0.318	0.273
	Britain	6	7	1	4	17	0.353	0.235
	Sweden	1	14	0	0	15	0.067	0.000

Het = heterozygous for the derived allele. Hom anc = homozygous ancestral. Hom der = homozygous derived. Other = missing or ambiguous genotypes, excluded from calculations. *n* = total called genotypes (Het + Hom anc + Hom der). Masked = fraction of called genotypes heterozygous for the derived allele (Het / *n*). Realised = fraction of called genotypes homozygous for the derived allele (Hom der / *n*).